

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations

April 2023

21BT304 - MOLECULAR BIOLOGY AND GENETIC ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Give a detailed account of "the central dogma".	10	L*1	2	1
	b) Explain in brief the mechanism of reverse transcription.	10	L1	2	2
2.	a) Briefly describe the process of DNA Replication in <i>E. coli</i> .	10	L1	2	1
	b) Write about the structure and function of DNA polymerase of prokaryotes and eukaryotes.	10	L1	2	2
3.	a) Explain in detail about chromosomes as carrier of genetic information.	10	L1	2	1
	b) Explain the structure and functions of DNA polymerases of prokaryotes.	10	L1	2	2
Unit – II					
4.	a) Discuss the process of maturation of functional proteins.	10	L1	2	3
	b) Discuss Bacterial RNA polymerase, its structure and function.	10	L1	2	4
5.	a) Elucidate different Post transcriptional modifications.	10	L1	2	3
	b) Explain the process of Translation in prokaryotes. State any four differences from eukaryotic translation.	10	L1	2	4
6.	a) Discuss the regulation of eukaryotic gene expression.	10	L1	2	3
	b) Elaborate Lac Operon as a model of gene regulation.	10	L1	2	4
Unit – III					
7.	a) Explain in brief different types of restriction enzymes.	10	L1	2	5
	b) Discuss in detail about the Vectors involved in recombinant DNA technology.	10	L1	2	5
8.	a) Explain Gene transfer techniques and electroporation.	10	L1	2	5
	b) Describe in detail about microprojectile system and liposomes.	10	L1	2	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
